Trigonometry & Polar Coordinates

Section 1: Product-to-Sum & Sum-to-Product Identities

1 Express $2\sin(5x)\cos(3x)$ as a sum of trigonometric functions.

a. $\sin(8x) + \sin(2x)$

b. $\sin(8x) - \sin(2x)$

c. $\cos(8x) + \cos(2x)$

d. $\cos(8x) - \cos(2x)$

2 Simplify the expression $\frac{\sin(3x) + \sin(x)}{\cos(3x) + \cos(x)}$.

a. $\tan(2x)$

b. $\tan(x)$

c. $\cot(2x)$

d. $\tan(4x)$

3 What is the exact value of $\cos(105^\circ) - \cos(15^\circ)$?

a. $-\frac{\sqrt{2}}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $-\frac{\sqrt{6}}{2}$

d. $\frac{\sqrt{6}}{2}$

Section 2: Polar Curves & Graphs

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4 What geometric curve is represented by the polar equation $r = 2 + 2\cos\theta$?

a. Cardioid

b. Limaçon with inner loop

c. Three-petal rose

d. Circle

5 How many petals does the rose curve $r = 4\cos(3\theta)$ have?

a. 3

b. 6

c. 4

d. 8

6 How many petals does the rose curve $r = 2\sin(4\theta)$ have?

a. 4

b. 8

c. 16

d. 2

Section 3: Polar to Rectangular Conversions

7 Convert the polar point $(4, \frac{5\pi}{6})$ to rectangular coordinates (x, y).

a. $(-2\sqrt{3}, 2)$

b. $(2\sqrt{3}, -2)$

c. $(-2, 2\sqrt{3})$

d. $(2, -2\sqrt{3})$

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8 Convert the rectangular point $(-3, -3)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(3\sqrt{2}, \frac{5\pi}{4})$

b. $(3\sqrt{2}, \frac{3\pi}{4})$

c. $(6, \frac{5\pi}{4})$

d. $(3\sqrt{2}, \frac{7\pi}{4})$

9 What is the Cartesian equation for the polar line $\theta = \frac{\pi}{3}$?

a. $y = \sqrt{3}x$

b. $y = \frac{\sqrt{3}}{3}x$

c. $y = x + \frac{\pi}{3}$

d. $x = \sqrt{3}y$

Section 4: Trigonometric Equations on Restricted Domains

10 Solve $\cos(3x) = \frac{1}{2}$ for $x \in [0, \pi]$.

a. $\frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{9}, \frac{7\pi}{9}$

d. $\frac{\pi}{3}, \frac{2\pi}{3}$

11 Find all solutions to $\sin(2x) = \cos(x)$ in the interval $[0, 2\pi)$.

a. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

b. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$

c. $\frac{\pi}{3}, \frac{2\pi}{3}$

d. $\frac{\pi}{6}, \frac{5\pi}{6}$

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12 What is the range of $f(x) = 2 - 3\sin(4x)$?

a. $[-1, 5]$

b. $[-3, 3]$

c. $[-5, 1]$

d. $[2, 5]$

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